Chapter 15 exercises • Security

OS15-E01 • Bounds and defense limits

A 12-byte C-string buffer receives 12 data bytes; no particular ABI layout is assumed.

1. How many bytes including NUL are needed? Give reject/truncate contracts.
2. Can NX prevent this overflow?

OS15-E02 • Pending-slot denial of service

There are 2000 slots, each incomplete request retained 8 s, without early cleanup.

1. Estimate the mean arrival rate filling them using one boundary.
2. Who can suffer from source-IP limits?

OS15-E03 • Reused keystream

m1=0x10, m2=0x70, s=0xA5, c=m XOR s.

1. Compute ciphertexts, their XOR, and recovery of m2 from known m1.
2. Does this require brute-force cipher breaking?

OS15-E04 • Non-coprime RSA plaintext

p=5, q=11, n=55, e=3, d=27, m=5.

1. Encrypt/decrypt with a square table.
2. Why does correctness hold despite gcd(5,55)≠1?

OS15-E05 • Hashes, MACs, and replay

Compare (m, H(m)), (m, MACk(m)), and a nonce-bound MAC.

1. Why does a public hash not authenticate, and a MAC not inherently prevent replay?
2. How do two threads observing a pending nonce avoid double acceptance?

OS15-E06 • Historical TLS identity and secrecy

A network adversary records a TLS1.2 RSA full handshake. Two years later the static RSA private key leaks.

1. Does a valid certificate signature alone prove current private-key possession?
2. Does deleting traffic keys necessarily protect the old session?

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Chapter 15 exercises • Security (continued)

OS15-E07 • Salt and guess cost

500 records have distinct salts, 20000 dictionary candidates each, and 2 ms/KDF on one serial work unit.

1. Give full-dictionary computation/time. Must salt be secret?
2. Does an online limit of one guess/account/s change offline time to 10000000 s?

OS15-E08 • IDS base rate

100000 events, attack fraction0.001, TPR0.9, FPR0.01.

1. Find expected TP, FN, FP, TN and precision.
2. At the same TPR/base rate, bound FPR for at least50% precision.

OS15-E09 • Firewall permissions

Use the chapter's first-match zones/rules: established, Internet→web443, web→DB5432, then deny.

1. Evaluate Internet→DB5432, web→DB5432, web→DB22, and an established DB→web reply.
2. Can a compromised web server still reach DB?

OS15-E10 • Windows 7 DACL and MIC

Use ordinary read/write masks without privilege/owner exceptions. U belongs to G and has low IL; object medium IL has NO_WRITE_UP.

1. Evaluate requests r and rw under [deny(G, w), allow(U, rw)].
2. Deliberately put allow first. What does DACL decide for rw, and what is the final write decision?
3. Distinguish NULL and empty DACLs.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study